



Estimation challenge

Implementation of KF and EKF

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1 3D position sensor raw measurements

Looking at the raw data of the 3D position sensor we can see that they are really noisy in respect to the ground truth data

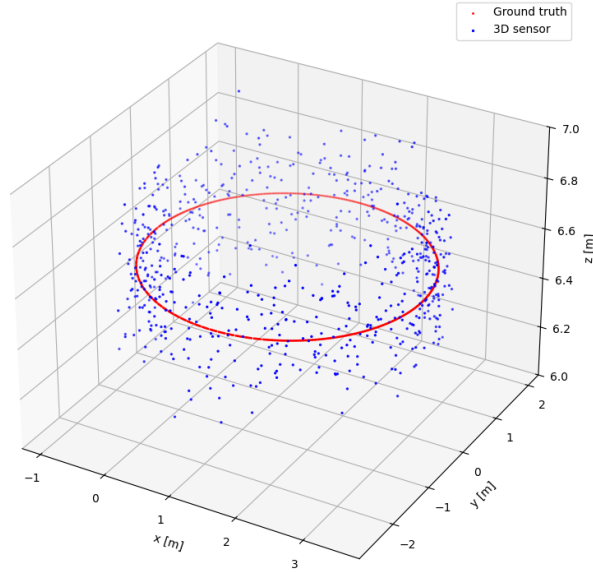


Figure 1: Raw position sensor data

Looking at the error along the 3 directions x, y, z we can compute the mean and the variance of the estimation error, as an additional metric to compare the different results is the RMS ATE [2] that represent the root mean square of the absolute trajectory error.

	Error x	Error y	Error z
Bias	-0.005 m	-0.005 m	-0.005 m
Variance	0.020 m^2	0.020 m^2	0.020 m^2
RMS ATE	0.246 m		

Table 1: Raw 3D data statistics

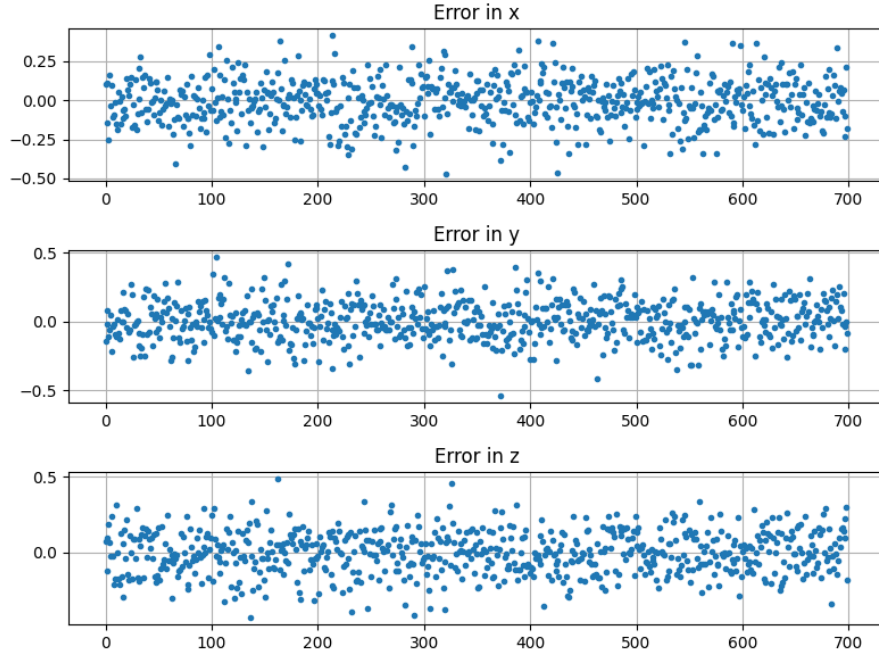


Figure 2: 3D sensor raw error

2 3D position sensor model

For this problem we can assume a constant velocity model to be able to perform a prediction step in the Kalman filter. The model of the system and of the measurement is:

$$\begin{cases} x_k = Ax_{k-1} + \epsilon_k \\ z_k = Hx_k + \eta_k \end{cases}$$

Where $\epsilon_k \sim N(\mathbf{0}, Q_k)$ and $\eta_k \sim N(\mathbf{0}, R_k)$. For a constant velocity model we consider as states both the positions and the velocities $x = (x, y, z, v_x, v_y, v_z)$, regarding the measurements with this sensor we have $z = (x, y, z)$. Therefore matrix H can be defined as:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

The measurement noise is defined in the task as

$$R = \begin{bmatrix} 0.02 & 0 & 0 \\ 0 & 0.02 & 0 \\ 0 & 0 & 0.02 \end{bmatrix} \text{m}^2$$

Since for a constant velocity model $x_k = x_{k-1} + v_x * \Delta t$ we can define

$$A = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

And for the process noise Q in the literature [1] is suggested

$$Q = \sigma_a^2 \begin{bmatrix} \frac{\Delta t^4}{4} & 0 & 0 & \frac{\Delta t^3}{2} & 0 & 0 \\ 0 & \frac{\Delta t^4}{4} & 0 & 0 & \frac{\Delta t^3}{2} & 0 \\ 0 & 0 & \frac{\Delta t^4}{4} & 0 & 0 & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & 0 & 0 & \Delta t^2 & 0 & 0 \\ 0 & \frac{\Delta t^3}{2} & 0 & 0 & \Delta t^2 & 0 \\ 0 & 0 & \frac{\Delta t^3}{2} & 0 & 0 & \Delta t^2 \end{bmatrix}$$

The process noise covariance Q accounts for modeling errors and unmodeled accelerations. Since the true motion is circular while the filter assumes constant velocity, Q is particularly important to compensate for the mismatch between the assumed and actual dynamics. The value of σ_a was tuned to reduce the error variance. An alternative way to find the best σ_a for this motion could be to perform a minimization comparing the estimated and real values to find the smallest variance value.

Since the measurement and system model are both linear we can use the Kalman filter algorithm defined as follows:

$$\begin{aligned} \text{Prediction step} & \begin{cases} \hat{x}_k^- = A\hat{x}_{k-1} \\ P_k^- = AP_{k-1}A^T + Q_k \end{cases} \\ \text{Update step} & \begin{cases} k_k = P_k^- H^T (HP_k^- H^T + R_k)^{-1} \\ \hat{x}_k = \hat{x}_k^- + K_k(Z_k - H\hat{x}_k^-) \\ P_k = (I - k_k H)P_k^- \end{cases} \end{aligned}$$

3 KF with 3D position sensor

3.1 Initial conditions A

The first Kalman filter was initialized with initial conditions $[-0.7, -0.3, 6.6, 0, 0, 0]$, we can then compare the results by looking at the 3d plot and at the error statistics.

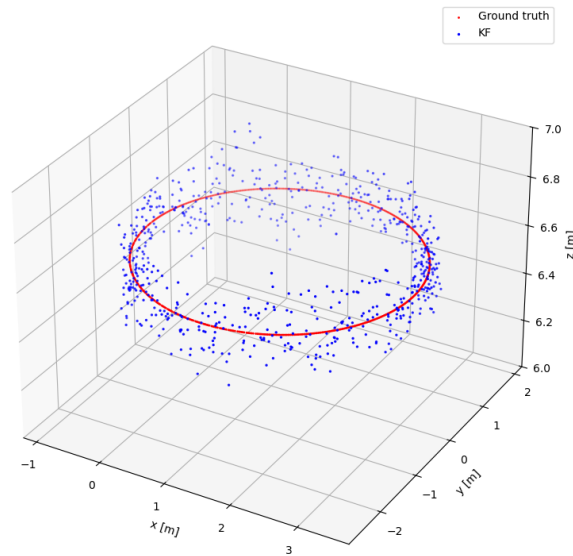


Figure 3: KF

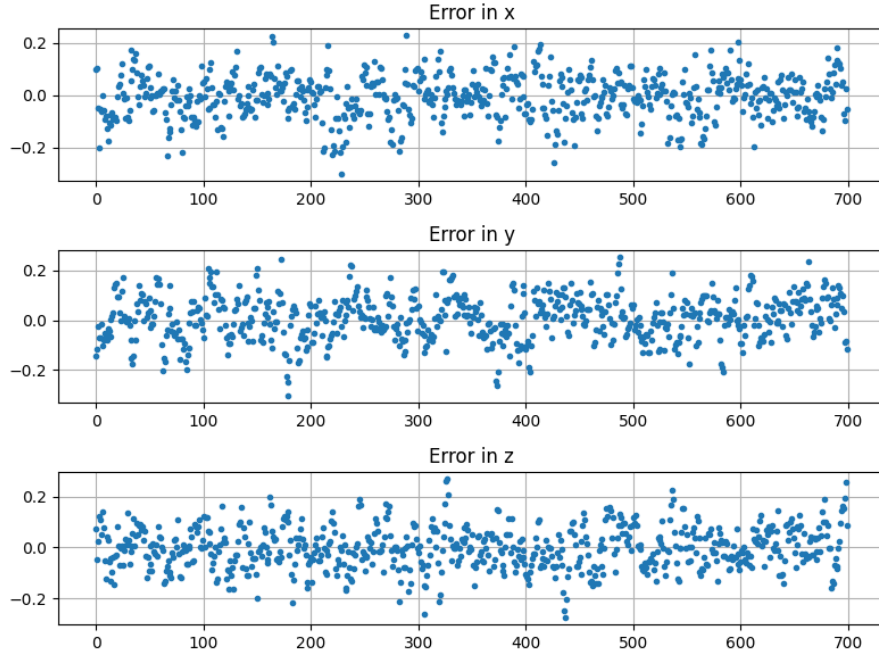


Figure 4: KF error

	Error x	Error y	Error z
Mean error	$-0.004m$	$0.009m$	$-0.002m$
Variance	$0.007m^2$	$0.008m^2$	$0.007m^2$
RMS ATE	$0.146 m$		

Table 2: KF estimation statistics

From this results we can see that the filter was able to improve the estimate. The error variance was reduced by approximately a factor of three.

3.2 Initial conditions B

Looking at the statistics when starting with the second set of initial conditions ($[-1.8, -0.5, 7.9, 0, 0, 0]$) we can see that the result is basically the same meaning that for this sensor the sensitivity to initial conditions is low. This can be an advantage because the filter is able to recover from wrong initial conditions.

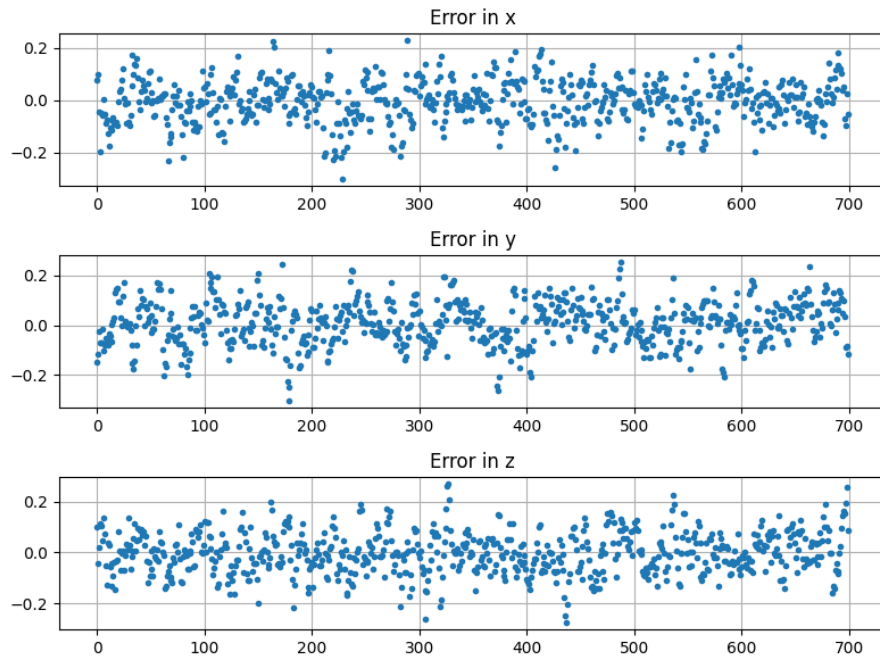


Figure 5: KF error (initial conditions B)

	Error x	Error y	Error z
Mean error	$-0.004m$	$0.009m$	$-0.002m$
Variance	$0.007m^2$	$0.008m^2$	$0.007m^2$
RMS ATE	$0.146 m$		

Table 3: KF estimate statistics (initial conditions B)

4 Camera raw measurements

The camera has a relatively small measurement noise in pixel coordinates but the measured quantity is not directly the position of the point and a transformation have to be applied.

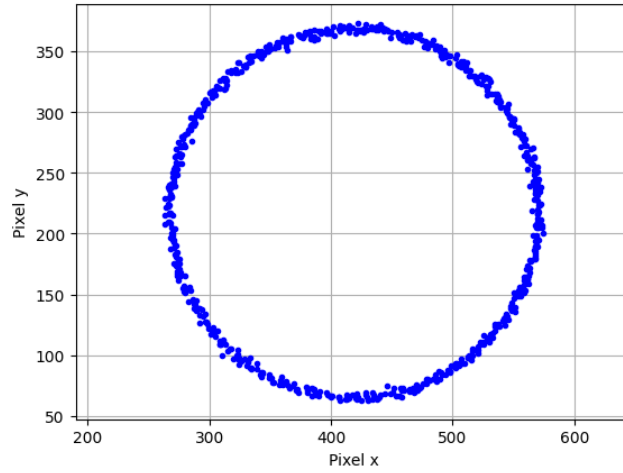


Figure 6: Camera reading

5 Camera model

From the pinhole camera model we can compute the relation between the 3D point to the coordinates on the camera sensor.

$$\begin{cases} u = \frac{f}{z}x + \frac{W}{2} \\ v = \frac{f}{z}y + \frac{H}{2} \end{cases}$$

This is the nonlinear system model $h(x)$ that can be linearized by computing the Jacobian

$$H = \frac{\partial h(x)}{\partial x} = \begin{bmatrix} \frac{f}{z} & 0 & -\frac{fx}{z^2} & 0 & 0 & 0 \\ 0 & \frac{f}{z} & -\frac{fy}{z^2} & 0 & 0 & 0 \end{bmatrix}$$

The measurement noise is again defined by the assignment and it is

$$R = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \text{px}^2$$

Since we have a nonlinear measurement system we need to use an Extended Kalman filter, the prediction step remains the same since the system model is linear and the update step becomes

$$\text{Update step} \begin{cases} k_k = P_k^- H^T (H P_k^- H^T + R_k)^{-1} \\ \hat{x}_k = \hat{x}_k^- + K_k (Z_k - h(\hat{x}_k^-)) \\ P_k = (I - k_k H) P_k^- \end{cases}$$

6 EKF with camera

6.1 Initial conditions A

Looking at the plot with the EKF estimated trajectory we can see an interesting problem.

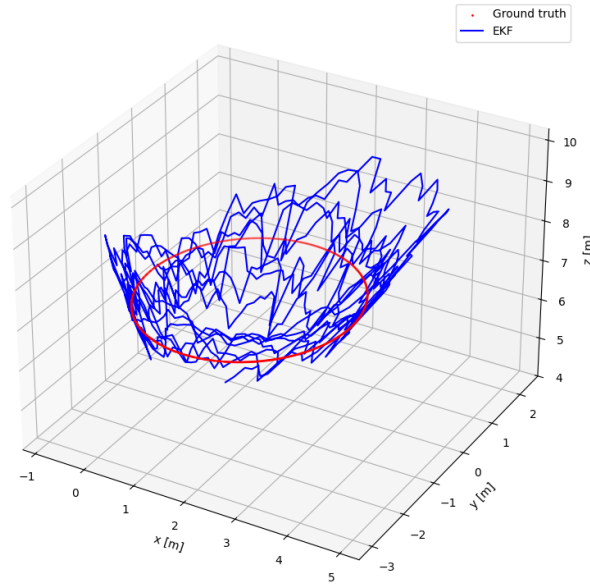


Figure 7: EKF 3D trajectory

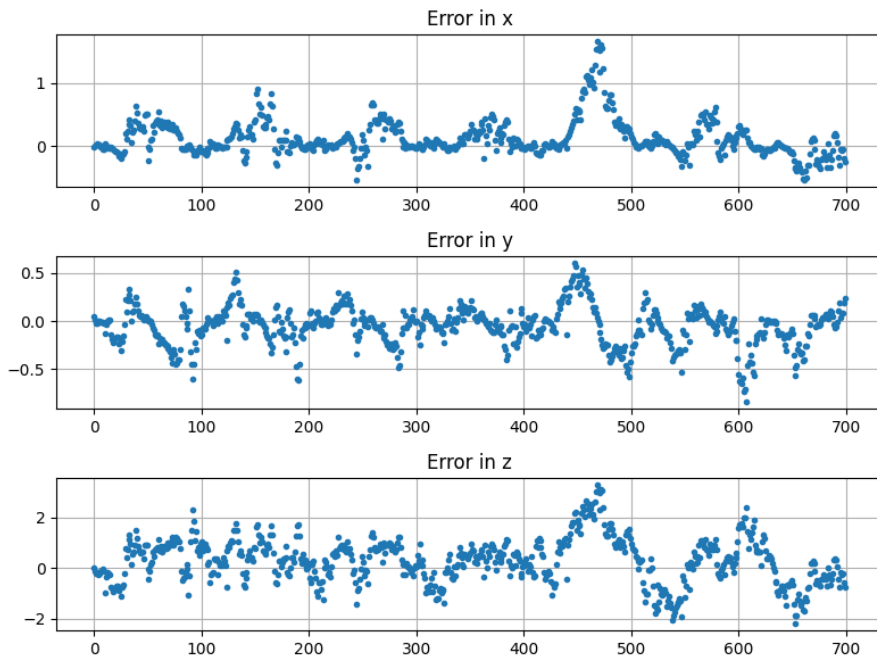


Figure 8: EKF error

Since the camera is only able to obtain a 2D measure we have no information about the distance of the point along z . The problem is related to the observability of the system. The camera measurement provides only the direction of the point with respect to the camera, while the absolute distance along that viewing ray is not directly observable.

Since every point along a line connecting the camera origin to the point could be a candidate estimate, the estimated state lies on a cone centered in the camera origin. For this reason, it is not possible to use the camera sensor alone but should be integrated with another sensor that can measure the distance of the point or the z coordinate.

Another alternative way of improving the result could be to maintain the z value fixed over time but that would lose the generality of the model used. This assumption can be adopted only if we have some knowledge about the trajectory of the point we want to measure or on

the dynamics of the object. The problem applying the assumption of constant z value would be that the estimate is really sensitive to initial conditions and we would have a constant bias along z .

6.2 Initial conditions B

The result with different initial conditions is not better and has still the same problems described above. The initial conditions matters only in case of a fixed z value.

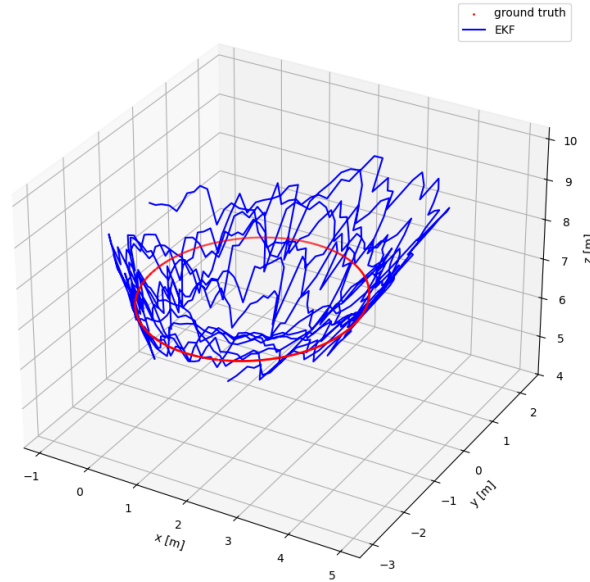


Figure 9: EKF 3D trajectory (initial conditions B)

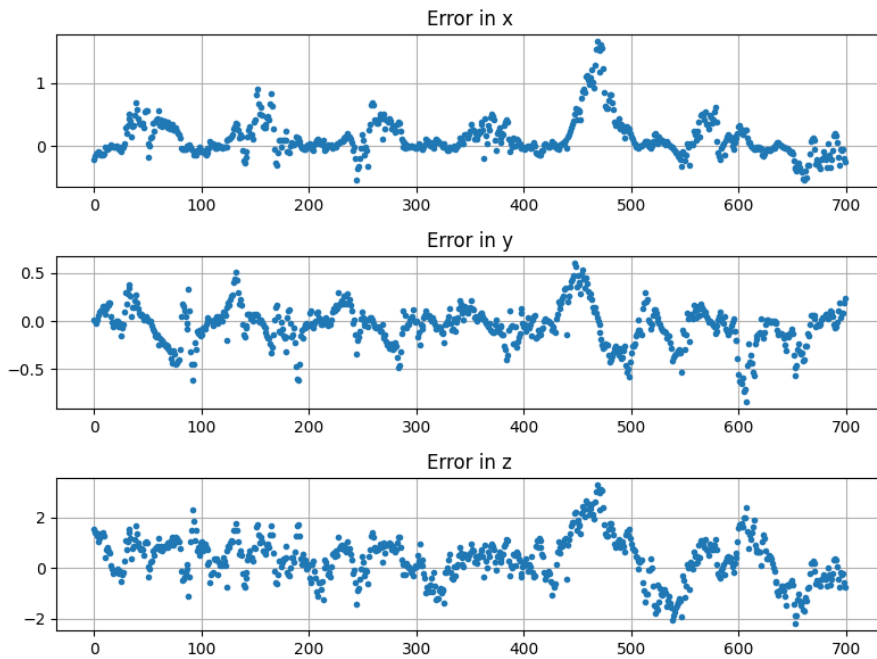


Figure 10: EKF error (initial conditions B)

7 Conclusions

The 3D sensor is very noisy but using a KF with a constant velocity model the noise could be reduced (from the result the final variance is about 1/3 of the original one), the advantage of this sensor is that it can measure all three position states of the point and so it has a little sensitivity to initial conditions. A way of improving the results further is to use a circular motion model instead of a constant velocity model but this would only work for this kind of motion and would lose generality.

A monocular camera does not directly provide absolute depth information. Therefore, the 3D position is not fully observable from a single camera measurement alone. Recovering the complete 3D trajectory requires additional information, such as a suitable dynamic model, prior knowledge about the motion, or measurements from an additional sensor.

The best solution would be to perform a sensor fusion between the camera and the 3D sensor, this can be done using again the EKF algorithm.

References

- [1] Gabriel F. Basso, Thulio Guilherme Silva De Amorim, Alisson V. Brito, and Tiago P. Nascimento. Kalman filter with dynamical setting of optimal process noise covariance. *IEEE Access*, 5:8385–8393, 2017.
- [2] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers. A benchmark for the evaluation of rgb-d slam systems. In *Proc. of the International Conference on Intelligent Robot Systems (IROS)*, Oct. 2012.